

Elias D. Crum

COMPUTER SCIENCE ENGINEERING PHD CANDIDATE · BIOINFORMATICIAN

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Education

Ghent University

Gent, Belgium

Doctor of Computer Science Engineering (in progress)

Oct. 2023 - Present

- KKnowledge on Web Scale (IDLab, UGent) / Digital Precision Health (VITO NV)

Loyola University Chicago

Chicago, IL, USA

Master of Science (Bioinformatics)

Aug. 2020 - Aug. 2022

- Graduated with Bioinformatics Honors
- GPA: 4.00 / 4.00

Loyola University Chicago

Chicago, IL, USA

Bachelor of Science

Aug. 2017 - May 2021

- Double Major: Bioinformatics & Biology
- Minor: History
- GPA: 3.991 / 4.000

Skills

Programming/Querying Languages: Python, BASH, R, Java, JavaScript, TypeScript, SPARQL, SHACL, HTML, CSS, Vue, LaTeX

Bioinformatics Tools: BowTie2, Kallisto, Spades, FastANI, PATRIC, PHASTER, Anvi'o, ResFinder, BLAST+, Clustal Omega, and related tooling

Miscellaneous: GitHub Actions, NGINX, Conda, Docker, Cytoscape

Semantic Web: SPARQL, RDF serialization, ontology definition, Solid

Languages: English (native), Dutch (A1), French (A2)

Honors and Awards

GRADUATE EDUCATION

Strategic Basic PhD Fellowship Awardee

2024-28

The Research Foundation - Flanders (FWO) · Brussels, Belgium

Four-year fellowship supporting privacy-aware clinical genomic data management and querying.

- Details: FWO Fellowship Record

Albare Award

2026

SWAT4HCLS 2026 · Amsterdam, The Netherlands

Poster award for VCF-to-RDF research, selected by consensus ranking across generative-AI models.

- Publication: From VCF to RDF: RML-Based Conversion Approaches for the Semantic Representation of Variant Data

Albare Award Certificate

Best Paper (Group)

2024

International Semantic Web Summer School 2024 · Bertinoro, Italy

Best scientific report award for the ISWS 2024 group research project.

- Slides: ISWS 2024 Group Project Presentation

Best Paper Certificate

Honors and Awards (continued)

Best Poster (Individual)

2024

International Semantic Web Summer School 2024 · Bertinoro, Italy

Best poster award for presenting the PENGQUIN clinical genomics research framework.

- Poster: ISWS 2024 Award-Winning Poster (PENGQUIN)

Best Poster Certificate

PRE-GRADUATE EDUCATION

Mulcahy Research Fellowship Award

2020-21

Recipient of a research scholarship for the 2020-2021 school year · Chicago, IL, USA

College of Arts and Sciences Dean's List Mention

2017-21

Eight-time Dean's List recipient · Chicago, IL, USA

Employee of the Year

2017-18

Campus Recreation, Loyola University Chicago · Chicago, IL, USA

Presidential Scholarship Recipient

2017-21

College of Arts and Sciences, Loyola University Chicago · Chicago, IL, USA

Research Experience

Graduate Researcher, Ghent University & VITO NV

Ghent, Belgium

Ph.D. Candidate

Oct. 2023 - Present

- Developing framework for clinical genomic data storage, sharing, and querying.
- Implementation of the framework utilizes the Solid protocol.
- Experience with SPARQL, RDF, ODRL, Linked Data, decentralized querying, and ontology definition.

SIB Swiss Institute of Bioinformatics

Lausanne, CH

Research Stay

Mar. - Apr. 2025 (2 months)

- Hosted by the Knowledge Representation Unit (KRU); worked directly with Jerven Bolleman, Tarcisio Mendes de Farias, Ana-Claudia Sima, and Vincent Emonet.
- Gained practical knowledge of public SPARQL endpoint maintenance.
- Developed Solid Cockpit to perform federated queries across public SPARQL endpoints and Solid pods containing research data, as part of the CHIST-ERA TRIPLE project.
- Funding: FWO Long Research Stay grant V416635N; project support through CHIST-ERA TRIPLE (CHIST-ERA-22-ORD-09).
- Direct contribution to: Does SPARQL federation work in the real world? A case study over large biological SPARQL endpoints (ISWC, Oct. 2026; accepted)
- Direct contribution to: Solid Cockpit: Data manager, privacy editor, and query facilitator for Solid pods (ESWC 2026 demo paper)

Master's Research, Loyola University Chicago (Putonti Lab)

Chicago, IL, USA

Master's Researcher

Aug. 2020 - Aug. 2022

- Identified and characterized novel, medically relevant bacteria and bacteriophage sequences.
- Master's thesis: Cataloguing Coliphages of the Human Urinary Microbiome.
- Experience with gene annotation and clustering, taxonomic classification screening, gene-similarity network production, genetic engineering and phage virulence augmentation.
- Publication: 10.1371/journal.pone.0283930

Undergraduate Research, Loyola University Chicago (Putonti Lab)

Chicago, IL, USA

Undergraduate Researcher

Aug. 2019 - Aug. 2020

- Performed independent research involving prokaryotic DNA sequencing, assembly, and analysis.
- Publication: 10.1128/mSphere.00154-21

Research Experience (continued)

Junior Researcher, Loyola Stritch School of Medicine (Wolfe Lab)

Chicago, IL, USA

Junior Research Assistant

May 2019 - Aug. 2019

- Researched the variable resistance of strains of bladder E. coli to bacteriophage infection.
- Presented the outcomes of this work to the Stritch School of Medicine Microbiology and Immunology Department.
- Presented a poster at the Stritch School of Medicine annual St. Albert's Day Research Celebration.

Professional Experience

Department of Biology, Loyola University Chicago

Chicago, IL, USA

Adjunct Professor of Biology

Aug. 2021 - May 2023

- Taught two sections of Biology Lab I and II each semester, with 24 students per section.
- Designed and delivered lectures, assessed in-class exercises, and developed and evaluated quizzes and exams.
- Collaborated with colleagues and students to maximize student learning and growth.

Emergency Department, University of Chicago Medical Center

Chicago, IL, USA

Medical Scribe

Nov. 2021 - Jan. 2023

- Directly assisted physician care of patients through patient intake charting, ordering tests, and other EPIC platform mediated physician logistical support.
- Interpreted and integrated medical charts to aid in the diagnosis and treatment of patients.
- Operated EPIC software for medical documentation.

Tutoring Center, Loyola University Chicago

Chicago, IL, USA

Training and Professional Development Committee Member

Nov. 2020 - May 2021

- Trained incoming student tutors and supplementary instructors.
- Supervised peer leaders and provided constructive feedback on instruction.
- Led group discussions and presented during bi-weekly meetings of more than 100 tutors.

Tutoring Center, Loyola University Chicago

Chicago, IL, USA

Supplementary Instructor (SI) and Group Tutor

Jan. 2019 - May 2021

- Tutored and served as professor's assistant for Pre-calculus (MATH 117) and Organic Chemistry I and II (CHEM 223 and 224).
- Attended lectures and aided professors with lessons for around 200 students.
- Hosted three tutoring sessions per week with an average attendance between 20 and 80 students.
- Tutored students in accelerated summer science courses including General Biology, Cell Biology, Genetics, General Chemistry, Organic Chemistry, Biochemistry, and Physics.

Campus Recreation, Loyola University Chicago

Chicago, IL, USA

Intramural Sports Official

Sept. 2017 - May 2019

- Refereed intramural sports including flag football, soccer, and volleyball.
- Communicated with other officials and players to ensure a fair and enjoyable intramural atmosphere.
- Diffused conflicts arising during or after games.
- Earned Campus Recreation Employee of the Year for the 2017-2018 academic year.

Professional Development

International Semantic Web Summer School 2024

Bertinoro, Italy

Participant

2024

- Participated in the 2024 International Semantic Web Summer School and related project activities.

Participation Certificate

Volunteer Experience

Inner City Education (ICE) Program

Ice Hockey and Skating Coach

Chicago, IL, USA

Oct. 2021 - Apr. 2023

- Coached underprivileged inner-city children in the fundamentals of skating and ice hockey.
- Taught, encouraged, and engaged with participants aged 8 to 15.

Alumni Relations, Loyola University Chicago

Student Alumni Ambassador

Chicago, IL, USA

Aug. 2019 - May 2020

- Attended Loyola alumni events and communicated the student experience with Loyola alumni.
- Worked at a diverse array of Loyola-sponsored events and acted as a representative of the university.

AMITA St. Joseph's Hospital

Emergency Room Volunteer

Chicago, IL, USA

Oct. 2018 - Mar. 2020

- Volunteered in the emergency room on a weekly basis.
- Helped restock rooms, tend to patient needs, and assist medical professionals.

Loyola4Chicago - Jordan Elementary

Afterschool Program Tutor

Chicago, IL, USA

Sept. 2019 - Mar. 2020

- Assisted in overseeing the afterschool homework program at Jordan Elementary for grades 3 and 4.
- Helped keep around 20 students focused while assisting with homework and occasional enrichment activities.

St. Paul Police PAL - Learn to Skate Program

Assistant Skating Instructor

St. Paul, MN, USA

Nov. 2015 - Mar. 2020

- Instructed underprivileged inner-city children in the fundamentals of skating and ice hockey.
- Assisted St. Paul Police officers with program coordination and participant organization.

Organizations

Alpha Sigma Nu Jesuit Honors Society Inductee

Loyola University Chicago Chapter · Chicago, IL, USA

2019

Delta Sigma Phi Fraternity

Epsilon Kappa Chapter, Loyola University Chicago · Chicago, IL, USA

2018

Publications

Does SPARQL federation work in the real world? A case study over large biological SPARQL endpoints

E. Crum, B.-E. Tam, J. Hanski, A.-C. Sima, T. Mendes de Farias, J. Bolleman, R. Taelman

International Semantic Web Conference (ISWC) 2026 · 2026 · Conference Paper · Accepted

Solid Cockpit: Data manager, privacy editor, and query facilitator for Solid pods

E. Crum, B. Buelens, G. Ertaylan, R. Taelman

European Semantic Web Conference (ESWC) 2026 · 2026 · Conference Paper

From VCF to RDF: RML-Based Conversion Approaches for the Semantic Representation of Variant Data

E. Crum, B. Buelens, G. Ertaylan, R. Taelman

SWAT4HCLS 2026 · 2026 · Poster Paper

Genomic sequence data sharing for clinical practice: A scoping review

E. Crum, R. Taelman, B. Buelens, G. Ertaylan

Computers in Biology and Medicine · 2026 · Journal Article

DOI: 10.1016/j.compbiomed.2025.111335

Publications (continued)

Observations on automated client-side query federation over Wikidata SPARQL endpoints

J. Hanski, E. Crum, R. Taelman

International Semantic Web Conference (ISWC) 2025 (WikiData workshop) · 2025 · Workshop Paper

Variant representation in RDF

N. Queralt-Rosinach, A. J. Kellmann, A. Bodrug-Schepers, A. Gaignard, E. Crum, P. Larmande, A. Waagmeester, J. Bolleman

BioHackrXiv · 2025 · Preprint

DOI: 10.37044/osf.io/jazsb_v1

Semantifying Genomic Variant Data: VCF to RDF Conversion Framework

E. Crum, R. Taelman, B. Buelens, G. Ertaylan, R. Verborgh

Proceedings of the 16th International SWAT4HCLS Conference (CEUR Workshop Proceedings) · 2025 · Conference Paper

Enriching Ontologies with Disjointness Axioms using Large Language Models

E. Crum, A. De Santis, M. Ovide, J. Pan, A. Pisu, N. Lazzari, S. Rudolph

International Semantic Web Conference (ISWC) 2024 (KBC-LM'24 workshop) · 2024 · Workshop Paper

DOI: 10.48550/arXiv.2410.03235

PerSoNal Genome QUery IN healthcare and clinical practice (PENGQUIN)

E. Crum

European Semantic Web Conference (ESWC) 2024 · 2024 · Conference Paper

DOI: 10.1007/978-3-031-78955-7_3

Personalized medicine through personal data pods

E. Crum, R. Taelman, B. Buelens, G. Ertaylan, R. Verborgh

Proceedings of the 15th International SWAT4HCLS Conference (CEUR Workshop Proceedings) · 2024 · Conference Paper

CHIST-ERA Triple: improving data interoperability and federation across RDF knowledge graphs and Solid Pods

J. Bolleman, E. Crum, I. Dragan, J. Galgonek, M. Ibberson, T. Mendes de Farias, M. Moos, M. Pagni, R. Taelman, J. Vondrášek, A. C. Sima

Proceedings of the 15th International SWAT4HCLS Conference (CEUR Workshop Proceedings) · 2024 · Conference Paper

Coliphages of the human urinary microbiota

E. Crum, Z. Merchant, A. Ene, T. Miller-Ensminger, G. Johnson, A. J. Wolfe, C. Putonti

PLoS ONE 18(4), e0283930 · 2023 · Journal Article

DOI: 10.1371/journal.pone.0283930

Cataloging and Engineering Temperate Coliphages of the Human Urinary Microbiome

E. Crum

Loyola University Chicago (Master's Thesis) · 2022 · Thesis

Genome Investigation of Urinary Gardnerella Strains and Their Relationship to Isolates of the Vaginal Microbiota

C. Putonti, K. Thomas-White, E. Crum, E. E. Hilt, T. K. Price, A. J. Wolfe

mSphere 6(3), e00154-21 · 2021 · Journal Article

DOI: 10.1128/mSphere.00154-21

ORCID: 0009-0005-3991-754X

Software & Semantic Resources

Solid Cockpit (github.com/KnowledgeOnWebScale/solid-cockpit)

Web application for interacting with Solid Pod resources in an integrated, user-friendly interface.

Active · updated Aug 2026 · Papers: ESWC 2026

Web Application

Software & Semantic Resources (continued)

Genome Sharing Paper Webpage (github.com/ecrum19/GenomeSharingPaper_Webpage)

Supporting website for the genomic sequence data sharing scoping review paper, intended to make national genome initiative data easier to browse, revisit, and update in the future.

Active · updated Mar 2026 · Papers: Computers in Biology and Medicine 2026

Companion Website

Federated Query Results Explorer (ecrum19.github.io/fed-survey-results)

Interactive results website for exploring longitudinal federation experiments over real-world biological SPARQL endpoints.

Activity not recorded · Papers: ISWC 2026

Companion Website

VCF-RDFizer (github.com/ecrum19/VCF-RDFizer)

A CLI tool for converting genomic variant data from VCF into RDF, supporting semantically interoperable downstream workflows.

Active · updated Sept 2026 · Papers: SWAT4HCLS 2026; BioHackrXiv 2025; SWAT4HCLS 2025

RDF Conversion Tool

Ontology Companion Generator (github.com/ecrum19/ocg)

An npm-based generator that creates GitHub Pages companions for exploring, visualizing, and documenting ontologies with configurable output.

Active · updated Sept 2026

Ontology Documentation Tool

<https://www.npmjs.com/package/ontology-companion-generator>

QR Code Generator (github.com/ecrum19/qr-code-generator)

Lightweight generator for producing QR codes that can be reused in posters, slides, and project materials.

Active · updated Aug 2026

Utility Tool

VCF-Core Vocabulary (github.com/ecrum19/vcf-core-vocabulary)

Vocabulary and SHACL shapes for representing the logical VCF 4.5 model in RDF.

Active · updated Sept 2026 · Papers: SWAT4HCLS 2026; BioHackrXiv 2025; SWAT4HCLS 2025

Semantic Vocabulary and SHACL Shapes

<https://ecrum19.github.io/vcf-core-vocabulary/>

Vocabulary of Restrictive Datasets (VoRD) (github.com/ecrum19/vord)

Semantic vocabulary resource for documenting and exploring VoRD in a browsable, companion-style format.

Active · updated May 2026

Semantic Vocabulary

SPARQL View Materialization Containers Specification (github.com/ecrum19/sparql-view-materialization-containers)

Container-based guidelines for reproducible SPARQL query view materialization for use in SPARQL query results provenance and caching.

No recent activity · updated Feb 2026

Technical Specification Draft

LDP Permissions Notifications Specification (github.com/ecrum19/ldp-permissions-notifications-specification)

Permissions and notification semantics technical specification for Linked Data Platform resources.

No recent activity · updated Feb 2026

Technical Specification Draft

Talks

Solid Cockpit

30/04/2026

Symposium · Solid Symposium

Demonstration of Solid Cockpit as a unified web interface for Solid Pods, integrating data management, privacy and access-control editing, notification support, and in-browser SPARQL data discovery.

Presentation slides

Genomic Data, Healthcare, and the Semantic Web

31/03/2026

Internal Meeting · KNoWS Day

KNoWS Day presentation outlining how semantic representations can improve genomic-data interoperability, privacy-aware access control, and federated querying in healthcare settings.

Presentation slides

18-Month Internal VITO PhD Jury

20/11/2025

Internal Meeting

Internal PhD milestone review covering completed work, validation status, and the next research phase.

Presentation slides

Digital Precision Health Day

10/10/2025

Internal Meeting

Presented progress on decentralized clinical-genomics workflows and interoperability priorities in precision health.

Presentation slides

SIB Research Stay Reflection

15/05/2025

Internal Meeting

Reflected on outcomes from the SIB research stay, including technical gains and collaboration opportunities.

Presentation slides

Swiss Public Health Network (SPHN)

29/04/2025

Research Stay Presentation

Introduced the PENGQUIN approach for privacy-aware clinical genomics integration using RDF and decentralized data pods.

Presentation slides

SIB Vital-IT

04/04/2025

Internal Meeting

Reviewed client-side federated SPARQL strategies for combining private Solid pod data with public knowledge graphs.

Presentation slides

SIB / SwissProt

25/03/2025

Research Stay Presentation

Presented semantic integration opportunities between curated protein resources and decentralized query workflows.

Presentation slides

TRIPLE Consortium: Intro to Solid

10/10/2024

Consortium Meeting

Introductory consortium session on Solid concepts, architecture, and relevance to TRIPLE project goals.

Presentation slides

FWO Interview

05/09/2024

Fellowship Interview

Funding interview pitch describing the PENGQUIN PhD vision, feasibility, and expected impact.

Presentation slides

Talks (continued)

International Semantic Web Summer School (ISWS) 2024: Group Project

15/06/2024

Summer School

Final group project presentation from ISWS 2024 highlighting design choices and implementation outcomes.

Presentation slides

ESWC - PhD Symposium

28/05/2024

Conference

PhD symposium talk on the PENGQUIN research agenda for decentralized and policy-aware genomic data use.

Presentation slides

Solid Symposium

02/04/2024

Symposium

Explored how Solid pods can host and govern personal genomic data for privacy-preserving analytics.

Presentation slides

KNoWS: An Intro to Genomics for Computer Scientists

04/03/2024

Internal Meeting

Foundational genomics overview tailored to computer science researchers entering health-data domains.

Presentation slides

Posters

Solid Cockpit - ESWC 2026 Poster

30/04/2026

Conference · ESWC

Poster overview of Solid Cockpit, a unified user interface for Solid Pods that combines data management, privacy control, notifications, and semantic querying.

Poster

SWAT4HCLS 2026 Poster: From VCF to RDF

24/03/2026

Conference · SWAT4HCLS

Poster presenting an RML-based method for converting full VCF variant files into RDF with stronger structure, provenance, interoperability, and privacy-aware semantics.

Poster

Semantifying Genomic Variant Data: VCF to RDF Conversion Framework

24/02/2025

Conference · SWAT4HCLS

Poster presenting a framework for semantically converting VCF genomic variant data into RDF using a dedicated ontology, HDT-based storage, and reusable mapping specifications.

Poster

PENGQUIN: Personal Genome Query in Healthcare and Clinical Practice

05/10/2024

Non-Academic Event · E(I)Xchange

Poster overview of the PENGQUIN research direction, focusing on privacy-aware storage, RDF representation, and distributed querying for personal genomic data.

Poster

PENGQUIN: Personal Genome Query in Healthcare and Clinical Practice

15/06/2024

Symposium · ISWS

Award-winning poster introducing the PENGQUIN research problem space around personal genomic data storage, privacy, semantic representation, and distributed querying.

Poster

Posters (continued)

PENGQUIN: Personal Genome Query in Healthcare and Clinical Practice

28/05/2024

Conference · ESWC

Conference poster summarizing the PENGQUIN approach for privacy-aware genomic data storage, RDF conversion, and patient-centric query workflows.

Poster

Personalized Medicine Through Personal Data Pods

26/02/2024

Conference · SWAT4HCLS

Poster describing how patient-controlled data pods could improve privacy, transparency, and data sharing for genomic medicine workflows.

Poster